

SEDAR 56 South Atlantic Black Seabass

- Establish a more comprehensive sampling program for ages and lengths of fish captured by the recreational fleet in all regions of the South Atlantic.
- Investigate discard mortality due to hooks in shallow waters (<10m)
- For this assessment, the age-dependent natural mortality rate was estimated by indirect methods. More direct methods, e.g. tag-recapture, might prove useful. Some tag-recapture studies have demonstrated relatively high tag return rates for black sea bass, at least compared to those of other reef fishes of the southeast U.S.
- Gather more depth data from private boat anglers.
- Investigate the potential for a range shift in the black sea bass population, and the potential causes, such as climate change.

The following are from SEDAR 25, and are still needed:

- The assessment panel recommended increasing the number of age samples collected from the general recreational sector.
- Black sea bass in the southeast U.S. were modeled in this assessment as a unit stock, as recommended by the DW and supported by genetic analysis (SEDAR 56-RD42). For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well as spatial patterns of recruitment. Even when fine-scale spatial structure exists, incorporating it into a model may or may not lead to better assessment results (e.g., greater precision, less bias). Spatial structure in a black sea bass assessment model might range from the very broad (e.g., a single Atlantic stock) to the very narrow (e.g., a connected network of meta-populations living on individual reefs). What is the optimal level of spatial structure to model in an assessment of snapper-grouper species such as black sea bass?
- The assessment time period (1978-2010) is short relative to some other assessments of South Atlantic reef fishes. Extending the assessment back in time might provide improved understanding of the stock's potential productivity and therefore sustainable yield, assuming the historic productivity is still relevant. Such an extension would require historic landings estimates from all fleets in operation. Although historic estimates from the commercial sector are available, those from the recreational sector are not. Hindcasting the historic recreational landings might require the development of new methods, or at least analysis of existing methods.
- Protogynous life history: 1) Investigate possible effects of hermaphroditism on the steepness parameter; 2) Investigate the sexual transition for temporal patterns, considering possible mechanistic explanations if any patterns are identified; 3)

Investigate methods for incorporating the dynamics of sexual transition in assessment models.

- In this assessment, the number of spawning events per mature female per year assumed a constant value of $X = 31$. That number was computed from the estimated spawning frequency and spawning season duration. If either of those characteristics depends on age or size, X would likely also depend on age or size. For black sea bass, does spawning frequency or spawning season duration (and therefore X) depend on age or size? Such dependence would have implications for estimating spawning potential as it relates to age structure in the stock assessment.
- For this assessment, the age-dependent natural mortality rate was estimated by indirect methods. More direct methods, e.g. tag-recapture, might prove useful. Some tag-recapture studies have demonstrated relatively high tag return rates for black sea bass, at least compared to those of other reef fishes of the southeast U.S.